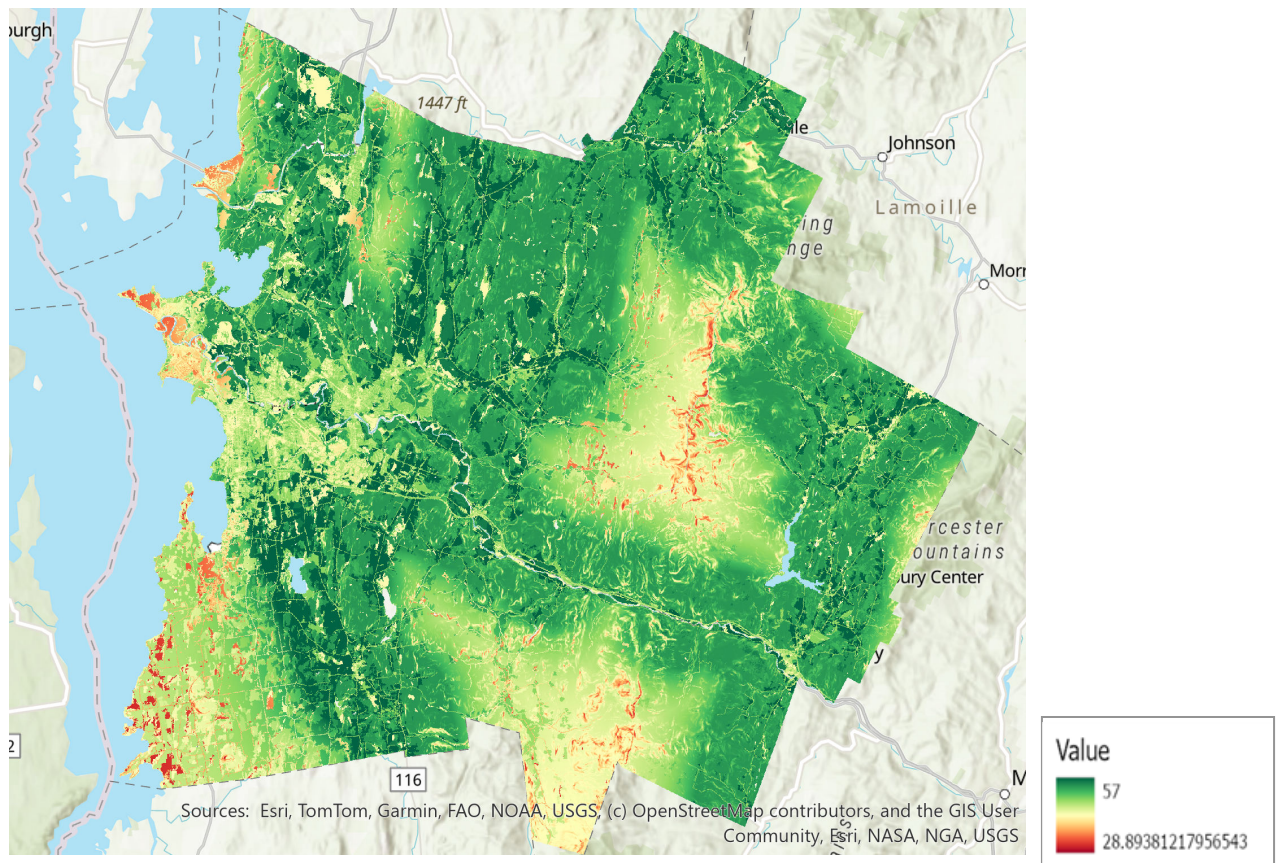


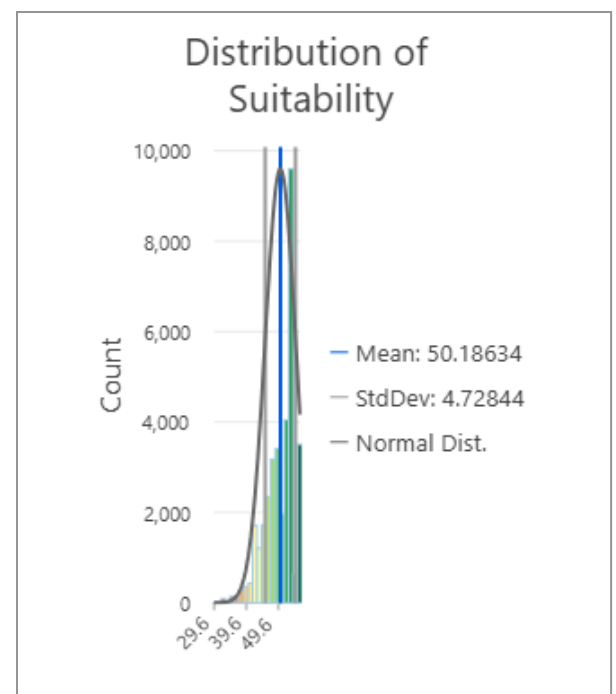


HQ_Building_Site Suitability model



Input criteria, preferences, and weights		
Criteria	Preferences	Weights
LandUse	Each value/category are assigned preferences.	1.6
Dist_MRoads	The smaller values have much higher preference.	1.25
Slope	The smaller values have higher preference.	1.6
Views_Zero	Small values most preferred and rapidly decreases.	1.25

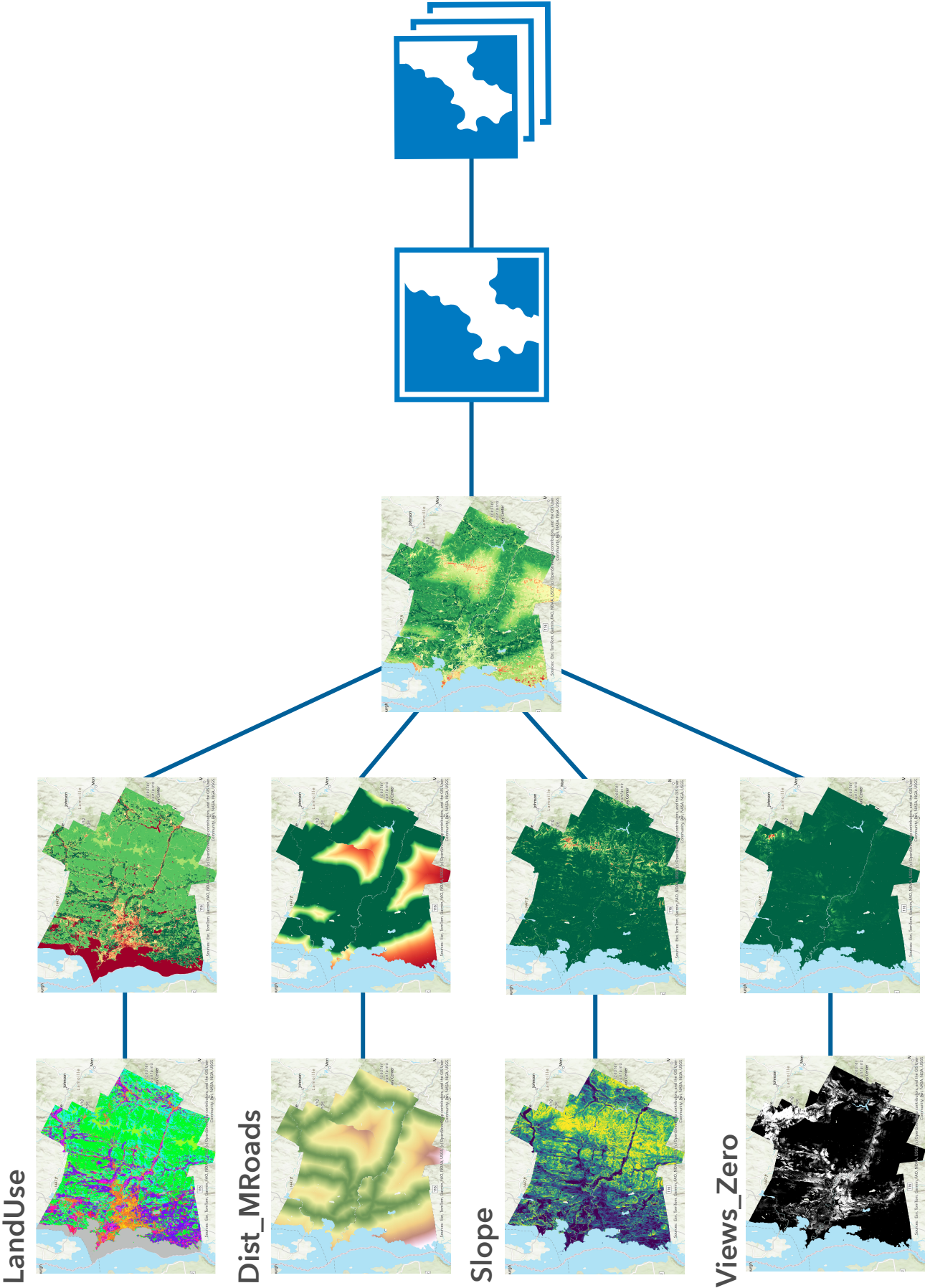
Suitability map statistics	
Minimum	28.894
Maximum	57
Mean	50.186
Cell size	30 Meter
Number of columns	2025
Number of rows	1858





Summary of modeling process

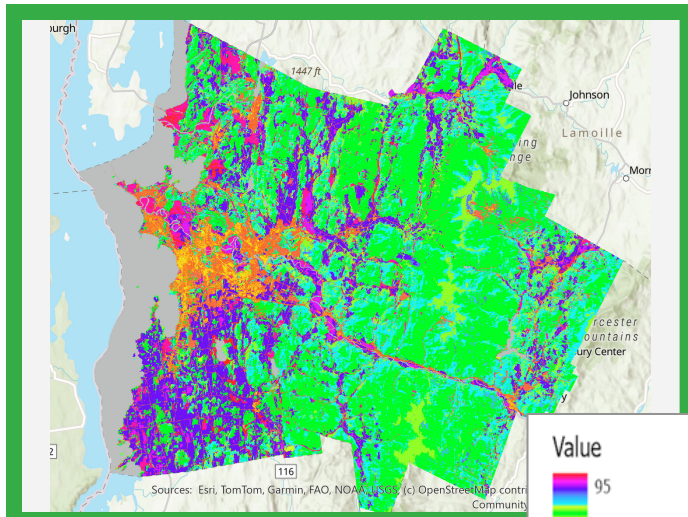
Model diagram





Summary of modeling process

Transforming – LandUse



Base criterion

The values in the input criteria have different meaning and ranges. Before they can be combined, they must be transformed onto a common suitability scale.

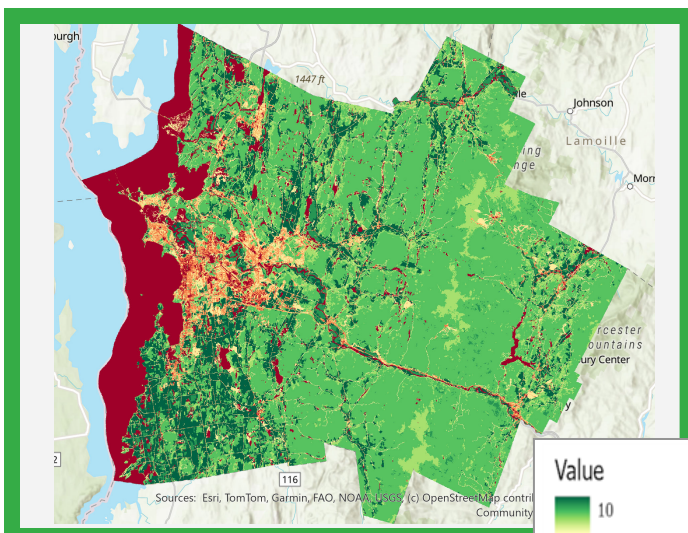
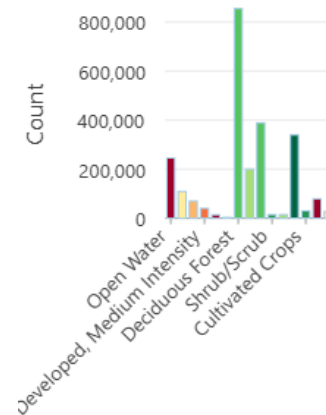
The values in this criterion represent unique categories. A one-to-one transformation was applied. Each category was assigned a suitability value, and is represented by a colored bar in the histogram. Only the first 10 categories are listed in the table.

Transforming criterion (1 of 4)

Transformation parameters

Open Water	1
Developed, Open	5
Developed, Low	4
Developed, Medium	3
Developed, High Intensity	1
Barren Land	8
Deciduous Forest	8
Evergreen Forest	7
Mixed Forest	8
Shrub/Scrub	9

Transformation of LandUse



Transformed criterion



Summary of modeling process

Transforming - Dist_MRoads

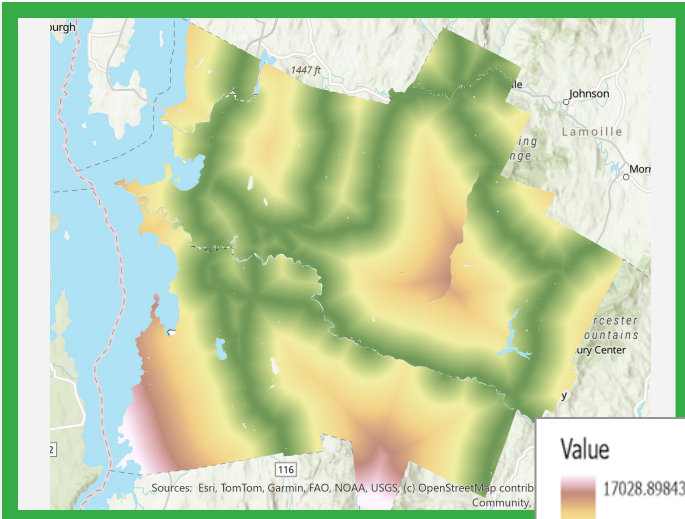
Base
criterion

Transform

Weight/
combine

Locate

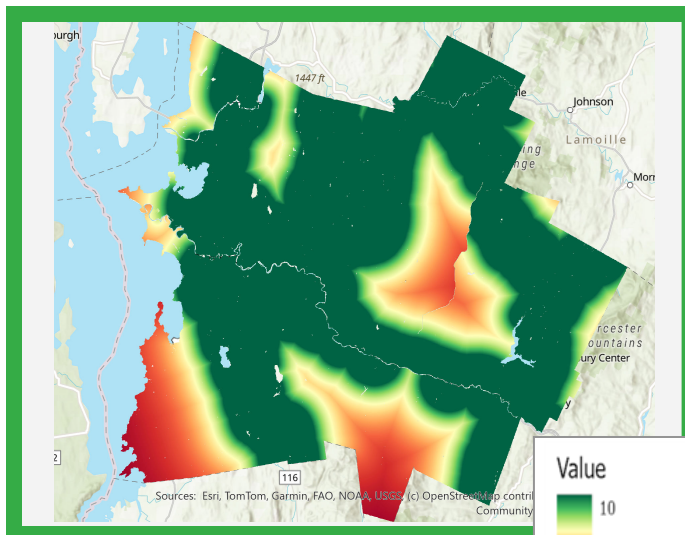
Evaluate



Base criterion

This criterion was transformed using the MSSmall continuous function (the blue line in the plot). The function is based on the mean and standard deviation of the input values.

Very small input values have higher preference. Locations that are closer to the features or lower in magnitude are more preferred.

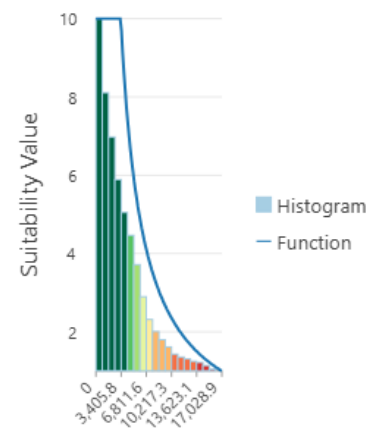


Transformed criterion

Transforming criterion (2 of 4)

Transformation parameters	
Type	Continuous
Function	MSSmall
Mean multiplier	1
Stdv multiplier	1
Lower threshold	0
Value below threshold	10
Upper threshold	17028.898
Value above threshold	1
Inverted	No

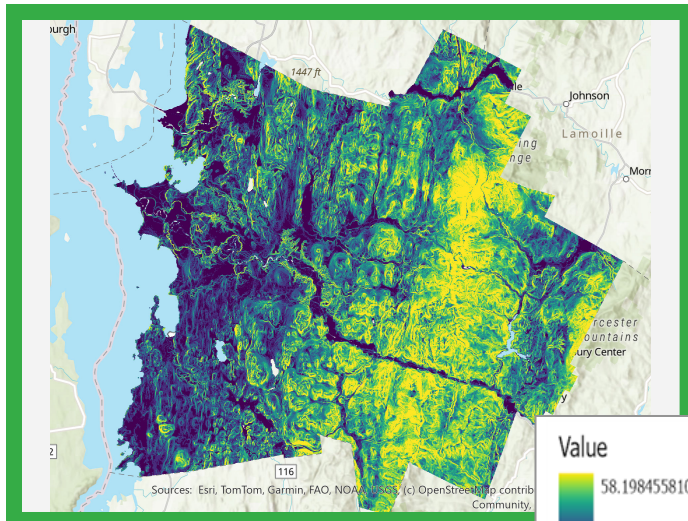
Transformation of Dist_MRoads





Summary of modeling process

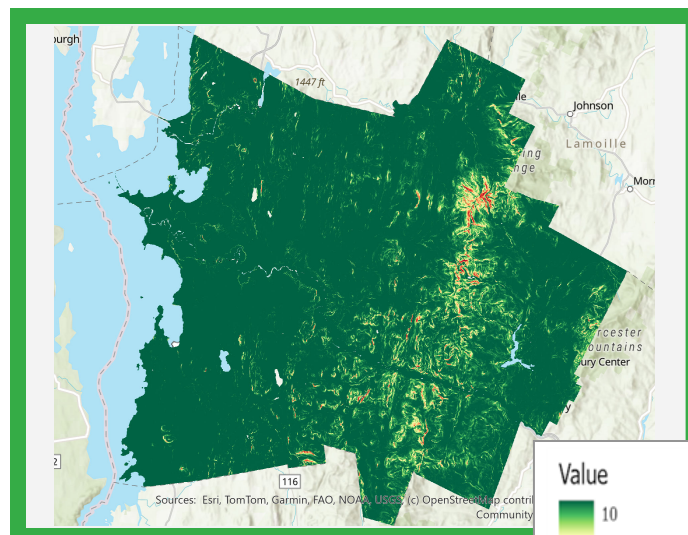
Transforming – Slope



Base criterion

This criterion was transformed using the Small continuous function (the blue line in the plot).

Smaller input values have higher preference. Locations closer to the features or lower in magnitude are more preferred. Values greater than the midpoint have decreased preference. Values below the midpoint have increased preference.

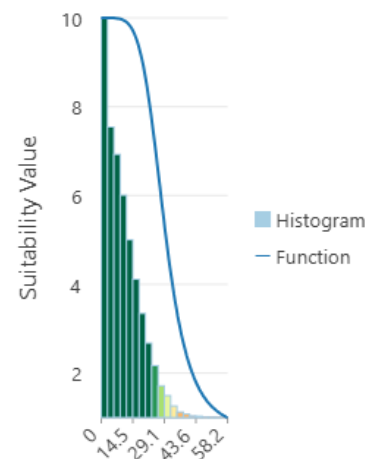


Transformed criterion

Transforming criterion (3 of 4)

Transformation parameters	
Type	Continuous
Function	Small
Mid point	29.099
Point spread	5
Lower threshold	0
Value below threshold	10
Upper threshold	58.198
Value above threshold	1
Inverted	No

Transformation of Slope





Summary of modeling process

Transforming - Views_Zero

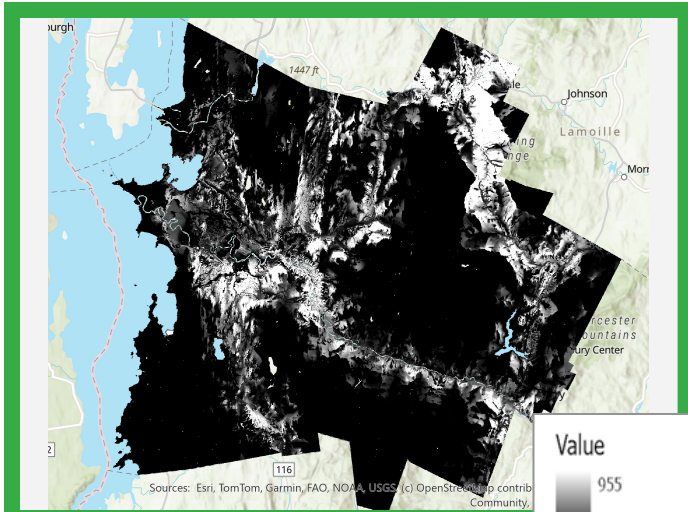
Base
criterion

Transform

Weight/
combine

Locate

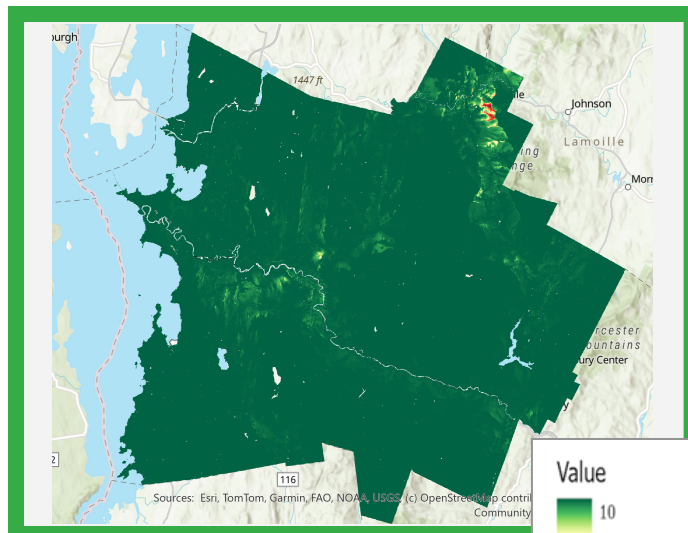
Evaluate



Base criterion

This criterion was transformed using the Logistic decay continuous function (the blue line in the plot).

The transformation fits a logistic decay function within a specified minimum and maximum using a defined y intercept percent. The small input values are most preferred. As the values increase, the preference rapidly decreases until they level off at the larger input values.

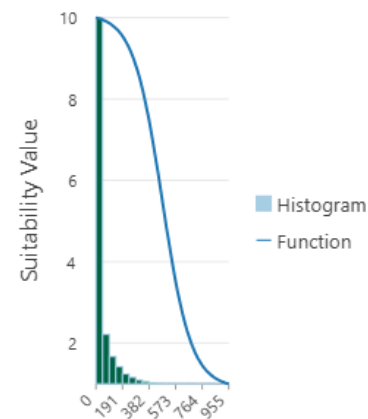


Transformed criterion

Transforming criterion (4 of 4)

Transformation parameters	
Type	Continuous
Function	LogisticDecay
Minimum value	0
Maximum value	955
Y-Intercept %	99
Lower threshold	0
Value below threshold	10
Upper threshold	955
Value above threshold	1
Inverted	No

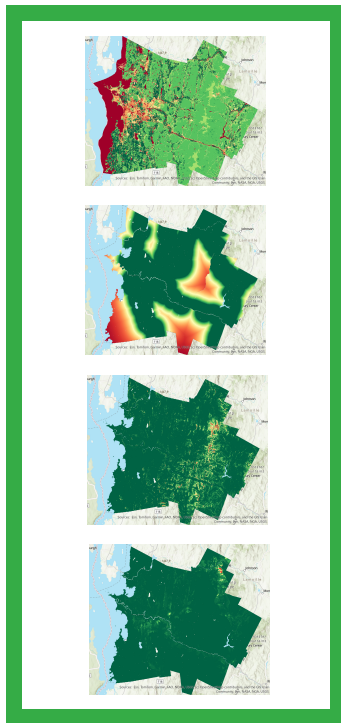
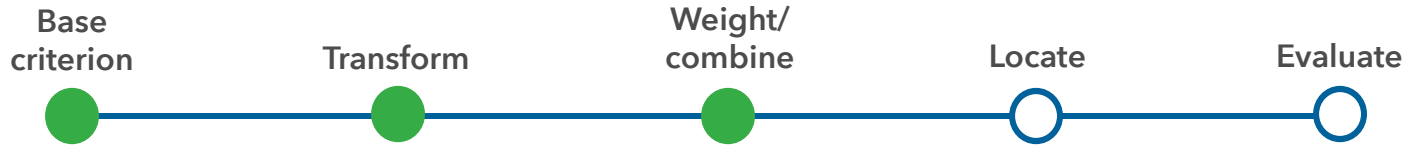
Transformation of Views_Zero



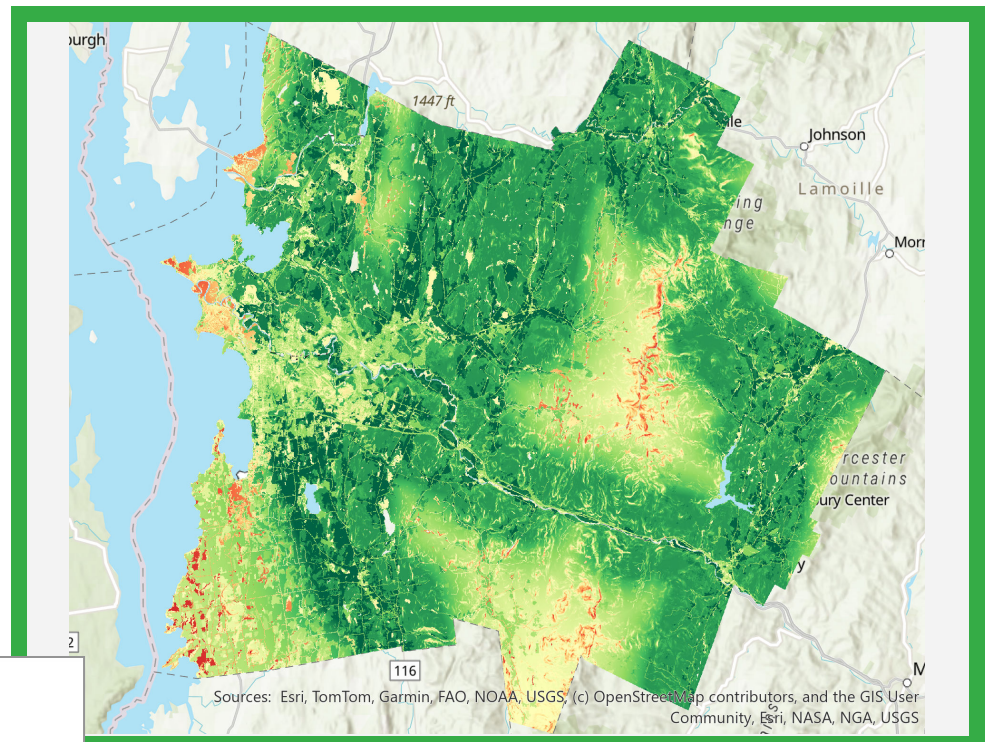


Summary of modeling process

Weight/combine – Suitability map



Criteria weights	
LandUse	1.6
Dist_MRoads	1.25
Slope	1.6
Views_Zero	1.25



This building site suitability submodel identifies locations that reduce construction cost and are easily accessible by employees. Locations assigned higher values are more preferred.